

The Erdős Institute
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Balancing Random Pokémon Battles

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Introduction

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- Pokémon is a turn-based game where players battle using creatures called Pokémon.
- Each choose a single action per turn in order to bring the opposing Pokémon's Hit Points to 0.



- Pokémon have various attributes which can be numeric or categorical.
- One popular format is “random Generation 9 battle” on the website Pokémon Showdown:
 - Randomly generated teams of 6;
 - Pokémon move sets are limited, and levels are scaled;
 - Pokémon aren't revealed to opponent or audience until fielded.

↑ A Random
Pokémon battle

Problem Statement

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Are randomly generated matches and teams “well-balanced” on Pokemon Showdown in regard to team statistics, types, and relative “advantages”? That is, at match-start, is the winner indeed a toss up?

Goal:

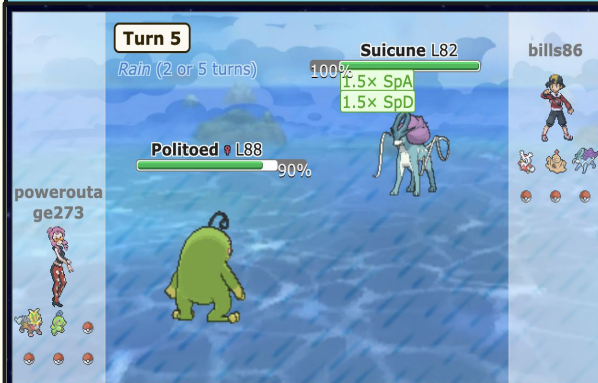
- Given the 12 Pokémon in a battle, predict the winner.

Stakes:

- Fair competition
- Player experience
- Platform credibility

Data Collection

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```
"id": "gen9randombattle-2646819945",
"format": "[Gen 9] Random Battle",
"players": [
  "poweroutage273",
  "bills86"
],
"log": "|j|\u2606poweroutage273\n|j|\u2606bills86\n|t:|
1783646439\n|gametype|singles\n|player|p1|poweroutage273
malva|1999\n|player|p2|bills86|101|1958\n|gen|9\n|tier|
[Gen 9] Random Battle\n|rated|\n|rule|Species Clause:
Limit one of each Pok\u00e9mon\n|rule|HP Percentage Mod:
HP is shown in percentages\n|rule|Sleep Clause Mod:
Limit one foe put to sleep\n|rule|Illusion Level Mod:
Illusion disguises the Pok\u00e9mon's true level\n|t:|
1783646439\n|teamsize|p1|6\n|teamsize|p2|6\n|start\n|
switch|p1a: Gouging Fire|Gouging Fire, L74|277/277\n|
switch|p2a: Delibird|Delibird, M|252/252\n|turn|1\n|
```

```
Code
1 from battle import *
2
3 replay = './gen9randombattle-2646819945.json'
4 bat = Battle(replay, parse=True)
5 bat.teams[0]
6
7 {'Gouging Fire': {'spec': Gouging Fire, lvl: 74, hp/m
8 'Politoed': {'spec': Politoed, lvl: 88, hp/max: 302/
9 'Meowscarada': {'spec': Meowscarada, lvl: 78, hp/max
10 'Forretress': {'spec': Forretress, lvl: 84, hp/max:
11 'Infernape': {'spec': Infernape, lvl: 82, hp/max: 25
12 'Gardevoir': {'spec': Gardevoir, lvl: 83, hp/max: 24
```

Random Battles

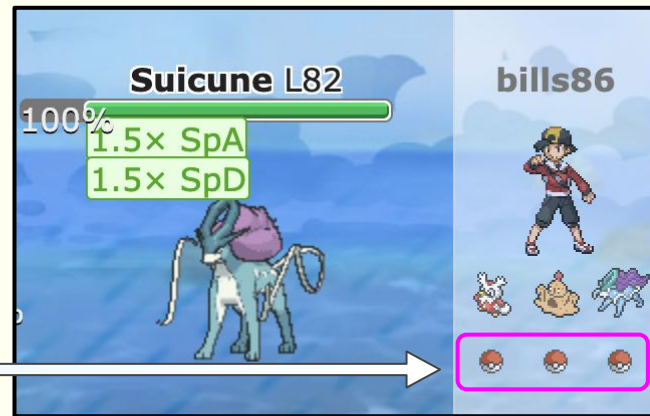
Battle logs

<Battle> class

A Problem!

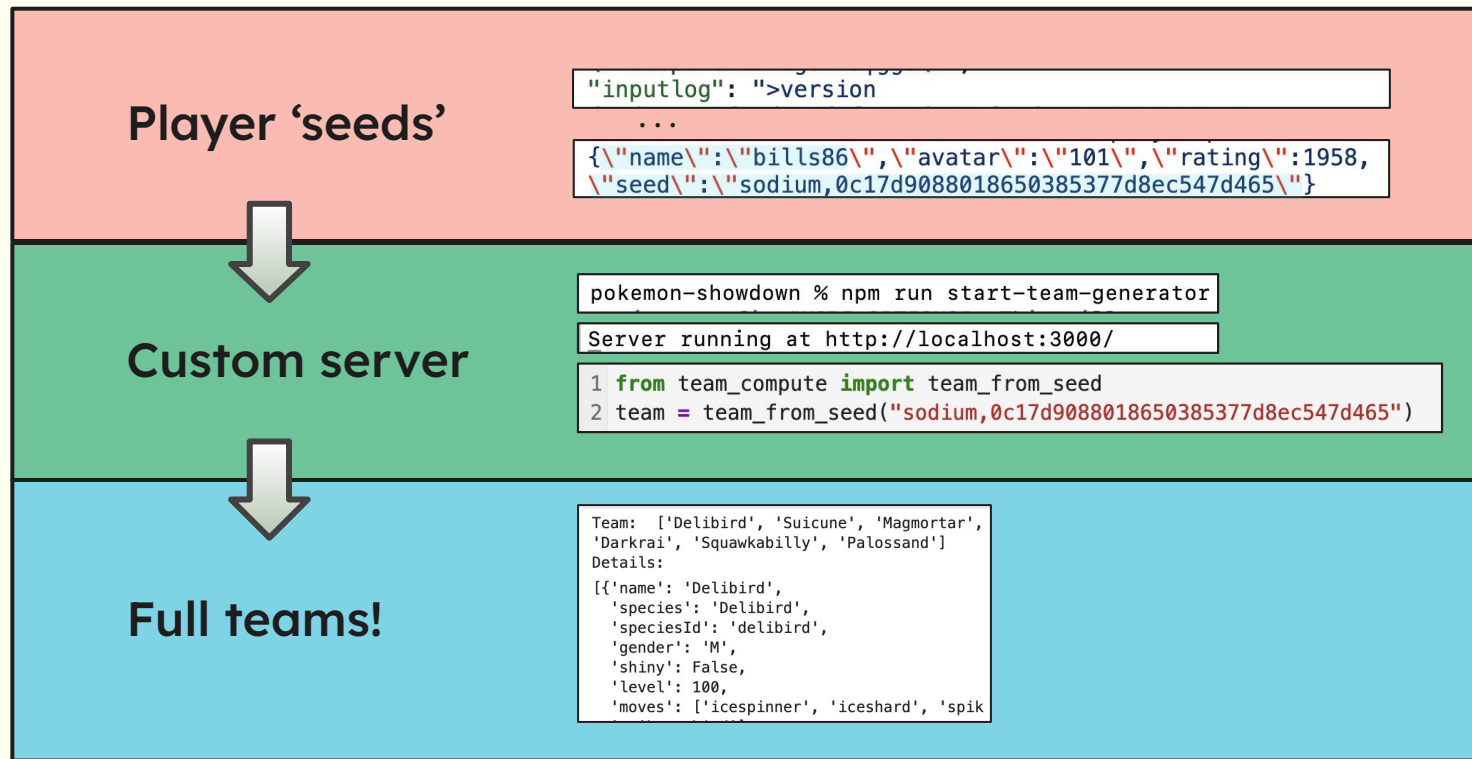
5

These Pokémon
aren't revealed (or
even logged) if not
used in the match!



The Fix

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Feature Engineering

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Overall team stats/characteristics:

- Speed, attack, defense, HP, Pokémon types.

Cross feature: Advantage stats

- Expectation of damage when Pokémon A attacks Pokémon B with some move.

Elo differences:

- Elo score of Player 1 minus Elo score of Player 2.

Speed difference:

- Pokémon with a higher speed moves first.

Type diversity:

- Counts of unique Pokémon types across teams.

Booster counts:

- Number of boosting moves and abilities.

Model Selection

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Model	Features	Cross-validation average accuracy	Well-calibrated?
Custom Baseline	Elo diff	52.1	Yes
Unbiased Log Reg	Elo diff, Advantage, Number of move + ability boosters	53.1	Yes
Biased Log Reg	All above + type diff + stat diff	54.1	No
Decision Tree	All above	53.3	No
Random Forest	All above	53.2	No
HGB	All above	52.0	No
XGB	All above	50.6	No

Conclusion

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- Final model: the unbiased logistic regression.
 - Calibration was an important deciding factor.
- Final model's accuracy on test data: 53.6%.
 - Compare baseline CV accuracy: 52.1%.
 - Compare final model's CV accuracy: 53.1%.

Conclusion: Gen-9 random battles on Pokémon Showdown are balanced!

Future Steps

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- Replace the damage approximation in the advantage statistic with an actual damage calculation.
- Construct turn-by-turn win probabilities.

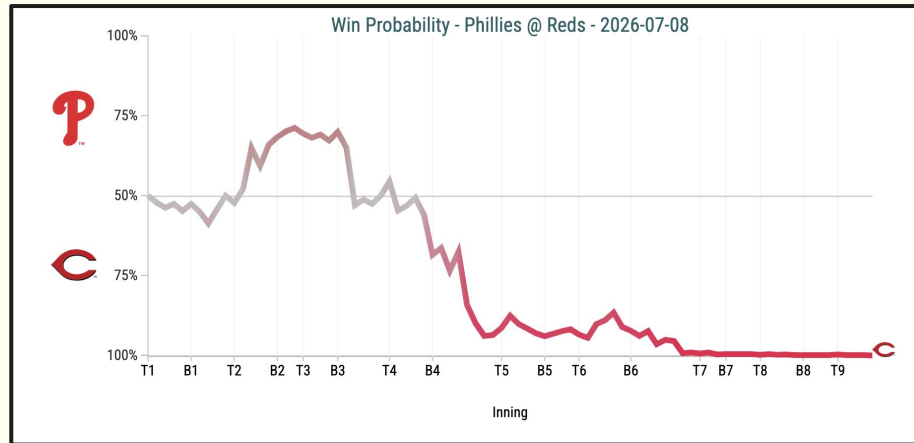


Image source: BaseballSavant



Thank You!